Frequency-Aware Processing for Subject-Invariant sEMG Gesture Recognition: From Signal Decomposition to Per-Band Style Mixing

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***Abstract*—Cross-subject generalization remains a key challenge in surface electromyography (sEMG) gesture recognition. We present a hypothesis-driven investigation (H1–H9) of *frequency-aware* processing for subject-invariant sEMG classification on NinaPro DB2 under strict leave-one-subject-out (LOSO) evaluation with 20 subjects and 10 gestures.**

Our spectral analysis (H1) reveals that inter-subject variability is frequency-dependent: normalized coefficient of variation grows from 0.20 in the 20–100 Hz motor-unit band to 2.04 above 700 Hz—a tenfold increase. Motivated by this finding, we show that $K = 4$ frequency-band decomposition via fixed Sinc filterbanks or learnable Unfolded VMD (UVMD) yields consistent improvements over raw-signal baselines (+3.7 pp in the controlled H6 ablation, $p_{\mathrm{Holm}} < 0.01$, Cohen's $d = 1.25$; also confirmed in H2, $p_{\mathrm{Holm}} < 0.02$). We further apply *per-band* MIXSTYLE—a zero-parameter style-augmentation module—which adds +1.1–1.8 pp on both decomposition types; on the learnable frontend (H7) this reaches significance within its hypothesis family ($p_{\mathrm{Holm}} < 0.01$), while on the fixed frontend the effect remains exploratory ($p_{\mathrm{uncorr.}} = 0.04$–$0.07$). In a descriptive experiment (H8), *Adaptive Per-Band* MIXSTYLE learns per-band parameters that correlate with the H1 gradient ($\alpha_4/\alpha_1 = 10\times$ on Sinc, Spearman $\rho = 0.78$, $p < 10^{-16}$), though this evidence is observational rather than based on controlled F1 comparisons.

We also propose *SCG-Net* (Spectral Content Gate Network, H9): a per-band learnable gate $\gamma_k \in [0, 1]$ controlling instance normalization strength. The gate learns a near-binary partition ($\gamma_4 = 0.998$ vs. $\gamma_{1\text{–}3} < 0.001$), and SCG-Net outperforms uniform Instance Normalization by +6.86 pp ($p_{\mathrm{Holm}} < 0.001$, 18/20 wins), though it does not exceed the MixStyle-based pipeline in absolute F1.

We report strong negative results: hard Instance Normalization degrades performance by -7 to -8.5 pp ($p_{\mathrm{Holm}} < 0.001$), and adversarial or MI-based disentanglement is harmful (-2.6 to -4.3 pp, $p_{\mathrm{Holm}} < 0.01$) under the architectures tested.

The combined pipeline achieves 37.7% macro-F1 (UVMD + per-band MixStyle), compared with the published LOSO baseline of 33.5%. All comparisons use Wilcoxon signed-rank tests with Holm–Bonferroni correction within pre-defined hypothesis families; we explicitly distinguish confirmatory and exploratory findings throughout.

***Index Terms*—sEMG, gesture recognition, cross-subject generalization, frequency decomposition, style augmentation, domain generalization**

I. Introduction

Surface electromyography (sEMG) enables non-invasive capture of voluntary muscle activation and is a key sensing modality for prosthetic hand control [1], rehabilitation [3], and human–computer interaction [4]. A critical obstacle to deployment is *inter-subject variability*: models trained on a pool of subjects degrade sharply when applied to a new, unseen user due to differences in electrode placement, skin impedance, subcutaneous fat, and individual motor-unit recruitment patterns [5].

Classical approaches to this problem include transfer learning [6], domain adaptation [7], and hand-crafted feature engineering with time-domain (TD) descriptors fed to Support Vector Machines (SVMs) [8]. More recently, deep learning architectures—convolutional neural networks (CNNs), gated recurrent units (GRUs), and temporal convolutional networks (TCNs)—have been applied directly to raw sEMG windows [1], [6], yet the cross-subject gap persists.

In this work we take a *frequency-aware* perspective. Our starting observation (Hypothesis H1) is that inter-subject spectral variability is not uniformly distributed across the sEMG bandwidth: high-frequency components exhibit up to ten times greater normalized variability than low-frequency motor-unit firing bands. This motivates a processing pipeline that *decomposes* the signal into sub-bands and applies *per-band style augmentation* rather than monolithic feature extraction.

We investigate this idea through a chain of nine hypotheses (table III), each building on the conclusions of the previous one, evaluated on the NinaPro DB2 dataset [2] under strict LOSO with up to 20 subjects and 10 gestures (Exercise E1, gesture IDs 8–17).

a) Contributions.:

1) A hypothesis-driven ablation study (H1–H9) on a unified experimental platform, quantifying the contribution of each processing component with paired statistical tests and Holm–Bonferroni correction within pre-defined hypothesis families.
2) Quantitative evidence that inter-subject sEMG variability is frequency-dependent (normalized CV ratio 10:1

between low and high bands), motivating per-band processing.

3) Confirmatory evidence that $K = 4$ frequency decomposition (fixed Sinc or learnable UVMD) is the largest contributor to cross-subject improvement in our pipeline (+3.7 pp, Cohen's $d = 1.25$, $p_{\mathrm{Holm}} < 0.01$).
4) Application of *per-band* MIXSTYLE—a zero-parameter style-augmentation module—to sEMG, yielding consistent gains (+1.1–1.8 pp) that are confirmatory on the learnable frontend ($p_{\mathrm{Holm}} < 0.01$) but remain exploratory on the fixed frontend.
5) *Adaptive Per-Band* MIXSTYLE (H8): a descriptive experiment showing that learned per-band parameters correlate with the H1 frequency gradient ($\alpha_4/\alpha_1 = 10\times$ on Sinc, Spearman $\rho = 0.78$, $p < 10^{-16}$), though the F1 effect of correct band ordering is small and not statistically confirmed.
6) *SCG-Net* (H9): a per-band *Spectral Content Gate* $\gamma_k \in [0, 1]$ that learns to selectively remove style. SCG-Net outperforms uniform Instance Normalization by +6.86 pp ($p_{\mathrm{Holm}} < 0.001$), though it does not exceed the MixStyle pipeline in absolute F1.
7) Negative results: adversarial and MI-based disentanglement degrades sEMG classification, and hard Instance Normalization is strongly harmful ($p_{\mathrm{Holm}} < 0.001$).

II. Related Work

A. Cross-Subject sEMG Recognition

The NinaPro database [2] is the de-facto benchmark for sEMG gesture recognition, providing standardized recordings from up to 67 subjects across multiple exercises. Pizzolato *et al.* [16] compared classical and deep classifiers on NinaPro, reporting that convolutional networks can match hand-crafted feature pipelines when sufficient training data is available. Golovan and Zvereva [9] recently established a reproducible LOSO baseline on DB2 with 20 subjects, reporting best macro-F1 of 33.5% (CNN/LSTM BiGRU on raw signals) and 32.6% (SVM-RBF on TD features), and demonstrating that intersubject variability—not model capacity—is the primary performance bottleneck.

a) Classical ML approaches.: Hudgins *et al.* [13] introduced the canonical set of time-domain (TD) features—mean absolute value, zero crossings, slope sign changes, and waveform length—for myoelectric pattern recognition, which remain widely used baselines. Phinyomark *et al.* [8] provided a systematic evaluation of EMG feature sets, showing that a small subset of TD features suffices for within-subject classification. Englehart and Hudgins [14] demonstrated that wavelet-based time-frequency features with linear discriminant analysis improve robustness over purely temporal representations. Scheme and Englehart [15] reviewed the gap between laboratory and clinical performance in myoelectric control, highlighting inter-session and inter-subject variability as the dominant unsolved challenges.

b) Deep learning on raw EMG.: Atzori *et al.* [1] applied CNNs directly to raw sEMG windows on NinaPro, establishing that deep learning can learn competitive representations without manual feature engineering. Côté-Allard *et al.* [6] proposed transfer learning with CNNs for inter-session sEMG, achieving improved robustness by pre-training on a source domain. Zhai *et al.* [17] introduced a self-recalibrating CNN framework that adapts to new subjects using minimal calibration data. Wei *et al.* [18] proposed multi-stream CNN architectures that process parallel temporal resolutions.

c) Transfer learning and domain adaptation.: Du *et al.* [7] applied domain adaptation via maximum mean discrepancy minimization to bridge the distribution shift between training and test subjects. Ketyko *et al.* [19] evaluated multiple domain adaptation strategies for cross-subject EMG, finding that simple fine-tuning often outperforms more complex adversarial methods. Campbell *et al.* [5] analyzed confounding factors—limb position, contraction intensity, and electrode shift—that compound the cross-subject challenge.

B. Frequency-Domain Processing for Biosignals

a) Empirical mode decomposition and VMD.: Huang *et al.* [20] introduced Empirical Mode Decomposition (EMD), which decomposes a signal into intrinsic mode functions adaptively, without requiring pre-specified basis functions. Rehman and Mandic [21] extended EMD to multivariate signals, enabling joint decomposition across multiple EMG channels. Dragomiretskiy and Zosso [10] proposed Variational Mode Decomposition (VMD), which recasts mode decomposition as a constrained optimization problem, yielding more robust sub-band separation than EMD.

b) Wavelet and time-frequency methods.: Englehart *et al.* [22] showed that discrete wavelet transform features outperform raw TD features for myoelectric classification by capturing transient frequency content. Phinyomark and Scheme [23] reviewed frequency-domain features for EMG, concluding that spectral features complement TD descriptors under non-stationary conditions.

c) Learnable filterbanks.: Ravanelli and Bengio [11] introduced parameterized Sinc filters (SINCNET) for speaker recognition, learning bandpass filter cutoffs end-to-end from raw waveforms. This approach has been adapted to EEG [24], demonstrating that learnable frequency decomposition can discover task-relevant sub-bands. We adapt this idea to sEMG and additionally propose an unfolded VMD (UVMD) alternative that inherits the optimization structure of VMD within a differentiable framework.

C. Style Transfer and Domain Generalization

a) Normalization and style manipulation.: Ioffe and Szegedy [25] introduced Batch Normalization; Ulyanov *et al.* [26] proposed Instance Normalization (IN), which normalizes each sample independently and removes style information in image generation. Huang and Belongie [27] demonstrated with Adaptive Instance

Normalization that feature statistics (mean and variance) encode "style," enabling arbitrary style transfer. These insights motivate our investigation of whether statistics-based normalization can remove subject-specific style in sEMG—our H3 results show that, unlike in vision, hard IN destroys gesture-relevant information.

b) Domain generalization.: Zhou *et al.* [12] proposed MIXSTYLE, a plug-in module that mixes feature statistics across source domains during training without requiring target domain data. We adapt MIXSTYLE to operate *per frequency band* in the sEMG setting. Sagawa *et al.* [28] introduced Group DRO, optimizing worst-group performance; Arjovsky *et al.* [29] proposed Invariant Risk Minimization (IRM). Ganin *et al.* [30] introduced the gradient reversal layer (GRL) for domain-adversarial training; our H4 experiments show this is counterproductive for sEMG. Wang *et al.* [31] surveyed domain generalization methods, distinguishing them from domain *adaptation* approaches [7], [19] that assume access to target data. Our work falls in the generalization setting: the test subject is never seen during training.

III. Dataset and Experimental Protocol

A. NinaPro DB2

We use the NinaPro Database 2 [2], comprising 40 intact-limb subjects performing hand gestures while 12 Delsys Trigno wireless electrodes record sEMG at 2000 Hz. Following the protocol of Golovan and Zvereva [9], we evaluate on Exercise E1 (gesture IDs 8–17, 10 grasping gestures, excluding rest), each repeated 6 times. We use a fixed subset of 20 subjects selected to span the full range of the database (subject IDs 1–5, 11–15, 26–30, 36–40), ensuring demographic diversity while keeping computational cost tractable.

B. Preprocessing

Raw sEMG signals are segmented into 200 ms windows (400 samples at 2000 Hz) with 50% overlap. Per-channel z-score standardization is applied using training-set statistics only (no test-set leakage). No bandpass filtering is applied; the decomposition frontend (Sinc or UVMD) serves as the learned filter.

C. LOSO Protocol

All experiments follow strict leave-one-subject-out (LOSO) cross-validation: for each fold, 19 subjects are used for training (with 10% held out for validation), and the remaining subject serves as the test set. We report macro-averaged F1 (primary) and accuracy (secondary), both averaged over 20 folds with standard deviation.

D. Statistical Testing

All pairwise comparisons use the Wilcoxon signed-rank test (two-sided) on paired per-subject F1 scores. We report Cohen's d (paired) as effect size, win/loss counts across subjects, and 10 000-resample bootstrap 95% confidence intervals for the mean F1 difference.

Table I: Inter-subject variability by frequency band (40 subjects). Normalized CV = CV of relative power (power in band / total power).

Band (Hz)	Raw CV	Norm. CV	Fisher Ratio
20–100	2.55	**0.20**	**0.079**
100–200	2.34	0.28	0.077
200–350	2.31	0.46	0.073
350–500	2.30	0.70	0.069
500–700	2.41	1.15	0.063
700–1000	2.44	**2.04**	**0.036**

a) Multiple-comparison correction.: Rather than applying a single Bonferroni correction across all comparisons ($k = 22$, overly conservative), we group tests into *hypothesis families* that share a common null and apply the Holm–Bonferroni step-down procedure [32] within each family:

- **Family A** ($k = 3$): Decomposition benefit (H2 Sinc, H2 UVMD, H6 A→B).
- **Family B** ($k = 3$): Soft style normalization (H3 MIXSTYLE, H6 B→C, H7 E→F).
- **Family C** ($k = 2$): Hard normalization harm (H3 Global IN, H3 Per-band IN).
- **Family D** ($k = 4$): Disentanglement losses (H4: contrastive, GRL, MI, combined).
- **Family E** ($k = 3$): SCG-Net variants (H9 C vs. D, A vs. D, C vs. A).

H5 uses a Friedman test (no pairwise correction needed); H8 comparisons are treated as descriptive (see section VI-H).

b) Confirmatory vs. exploratory.: We distinguish **confirmatory** results ($p_{\mathrm{Holm}} < 0.05$) from **exploratory** results ($p_{\mathrm{uncorr.}} < 0.05$ but $p_{\mathrm{Holm}} \geq 0.05$). Throughout the paper, p-values are labeled as p_{Holm} (corrected within family) or $p_{\mathrm{uncorr.}}$ to avoid ambiguity.

IV. Spectral Analysis of Inter-Subject Variability

To motivate frequency-aware processing, we first characterize how inter-subject variability distributes across the sEMG spectrum. For each of 40 subjects, we compute the power spectral density (PSD) via Welch's method (1024-point FFT, 50% overlap) and partition the Nyquist bandwidth into six bands: 20–100, 100–200, 200–350, 350–500, 500–700, and 700–1000 Hz.

A. Coefficient of Variation

For each band, we compute the coefficient of variation (CV) of relative power across subjects, averaged over gestures and channels. Table I shows that *normalized* CV increases monotonically from 0.20 in the motor-unit firing band (20–100 Hz) to 2.04 above 700 Hz—a **tenfold ratio**. Raw (unnormalized) CV is relatively flat (∼2.3–2.6), indicating that the variability pattern emerges only when power distribution is considered.

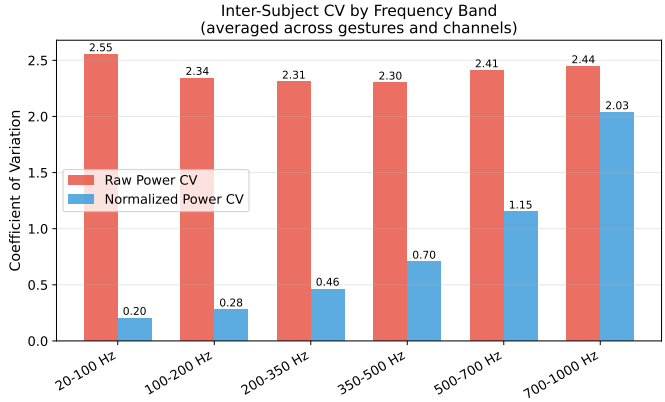

Figure 1: Inter-subject coefficient of variation (CV) by frequency band, averaged over 10 gestures and 12 channels (40 subjects). Red: raw power CV (flat ∼2.3–2.6). Blue: normalized CV (power in band / total power), showing a **tenfold increase** from 20–100 Hz to 700–1000 Hz.

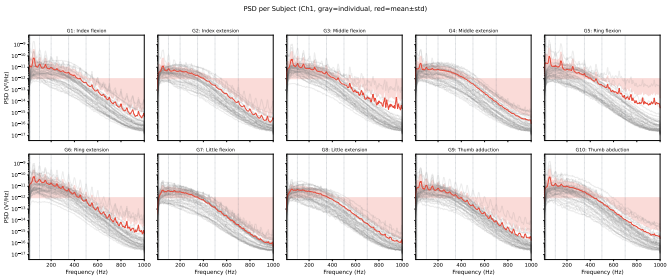

Figure 2: Welch PSD per gesture (Channel 1, 40 subjects). Gray lines: individual subjects; red: mean ± std. The inter-subject envelope widens non-uniformly across the spectrum, motivating per-band processing.

B. Fisher Discriminant Ratio

The Fisher ratio (between-gesture variance / total subject variance) decreases from 0.079 (20–100 Hz) to 0.036 (700–1000 Hz), confirming that high-frequency bands are both *more variable across subjects* and *less discriminative between gestures* (table I).

Figure 1 visualizes this gradient: after normalizing by total power, the inter-subject CV increases monotonically from the motor-unit band to the high-frequency tail, while raw CV remains flat. Figure 2 shows the underlying cause: PSD overlays for each gesture reveal that the "fan" of individual subject curves widens non-uniformly across the spectrum, with the greatest spread at higher frequencies.

Implication. A monolithic model that processes all frequencies identically must trade off suppression of high-frequency subject noise against preservation of low-frequency gesture information. Frequency decomposition allows per-band processing, applying stronger normalization where variability is highest.

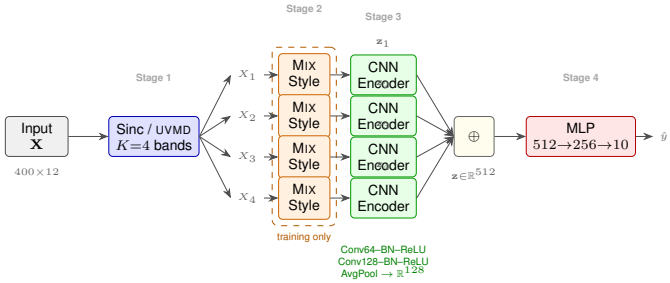

Figure 3: Architecture of the proposed frequency-aware sEMG pipeline. Stage 1 decomposes the raw input into $K{=}4$ frequency sub-bands via either a fixed Sinc filterbank or learnable UVMD. Stage 2 applies per-band MIXSTYLE (training only; zero parameters) to mix channel-wise statistics between subjects. Stage 3 encodes each sub-band independently through shared-architecture CNN encoders, producing $\mathbf{z}_k \in \mathbb{R}^{128}$. The concatenated representation $\mathbf{z} \in \mathbb{R}^{512}$ is classified by an MLP head (Stage 4).

V. Proposed Method

Our pipeline (fig. 3) processes a raw sEMG window $\mathbf{X} \in \mathbb{R}^{T\times C}$ ($T = 400$ samples, $C = 12$ channels) in four stages:

A. Stage 1: Frequency Decomposition

We decompose $\mathbf{X}$ into $K = 4$ sub-band signals $\{\mathbf{X}_k\}_{k=1}^{K}$, each $\in \mathbb{R}^{T\times C}$, using one of two frontends:

a) Fixed Sinc Filterbank.: Four bandpass Sinc filters with fixed cutoff frequencies partition the Nyquist band into equal sub-bands. For filter k with passband $[f_k^{\text{lo}}, f_k^{\text{hi}}]$:

$$h_k[n] = 2f_k^{\text{hi}}\,\text{sinc}(2f_k^{\text{hi}}n) - 2f_k^{\text{lo}}\,\text{sinc}(2f_k^{\text{lo}}n) \tag{1}$$

windowed by a Hamming function, applied via convolution to each channel independently. The cutoffs are fixed at $[0, 0.125, 0.25, 0.375, 0.5]$ (normalized frequency), corresponding to four equal-width bands. This frontend adds only 8 parameters (4 pairs of cutoff frequencies).

*b) Unfolded VMD (*UVMD*).:* We *unroll* $L = 8$ iterations of the alternating direction method of multipliers (ADMM) update for Variational Mode Decomposition [10] into a differentiable neural network block. The learnable parameters are:

- $\boldsymbol{\alpha} \in \mathbb{R}^{L\times K}$: per-iteration, per-mode bandwidth penalty,
- $\boldsymbol{\tau} \in \mathbb{R}^{L}$: per-iteration Lagrangian step size,
- $\boldsymbol{\omega} \in \mathbb{R}^{K}$: center frequencies (initialized uniformly at $[0.05, 0.18, 0.32, 0.45]$).

At each ADMM iteration l and for each mode k, the spectral-domain update reads:

$$\hat{u}_k^{(l)}(\xi) = \frac{\hat{r}_k(\xi)}{1 + \alpha_{l,k}(\xi - \omega_k)^2} \tag{2}$$

where $\hat{r}_k$ is the residual after removing all other modes. A spectral overlap penalty $\mathcal{L}_{\text{overlap}} = \sum_{j\neq k} \exp(-10(\omega_j - \omega_k)^2)$ encourages frequency diversity. This frontend adds 44 parameters ($\boldsymbol{\alpha}$: 32, $\boldsymbol{\tau}$: 8, $\boldsymbol{\omega}$: 4).

B. Stage 2: Per-Band Encoder

Each sub-band $\mathbf{X}_k$ is independently encoded by a shared-architecture 1D-CNN:

$$\mathbf{z}_k = \mathrm{AvgPool}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}_{128}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}_{64}(\mathbf{X}_k)\big)\big)\big)\big)\big)\big) \quad (3)$$

where Conv_d denotes $\mathrm{Conv1d}(C, d, \cdot)$ with kernel sizes 5 and 3 respectively, and AvgPool is global adaptive average pooling. Each encoder maps $\mathbb{R}^{T\times C} \to \mathbb{R}^{128}$; four encoders yield $\mathbf{z} = [\mathbf{z}_1; \ldots; \mathbf{z}_4] \in \mathbb{R}^{512}$.

C. Stage 3: Per-Band MixStyle

During training only, we apply MixStyle [12] independently to each sub-band *before* encoding. For sub-band k and sample i in a mini-batch:

$$\mu_k^{(i)} = \mathbb{E}_t[\mathbf{X}_k^{(i)}], \quad \sigma_k^{(i)} = \sqrt{\mathrm{Var}_t[\mathbf{X}_k^{(i)}] + \epsilon} \quad (4)$$

$$\tilde{\mathbf{X}}_k^{(i)} = \frac{\mathbf{X}_k^{(i)} - \mu_k^{(i)}}{\sigma_k^{(i)}} \quad (5)$$

$$\lambda \sim \mathrm{Beta}(\alpha, \alpha), \quad j = \mathrm{shuffle}(i) \quad (6)$$

$$\tilde{\mu}_k = \lambda\mu_k^{(i)} + (1-\lambda)\mu_k^{(j)}, \quad \tilde{\sigma}_k = \lambda\sigma_k^{(i)} + (1-\lambda)\sigma_k^{(j)} \quad (7)$$

$$\hat{\mathbf{X}}_k^{(i)} = \tilde{\sigma}_k \cdot \tilde{\mathbf{X}}_k^{(i)} + \tilde{\mu}_k \quad (8)$$

with $\alpha = 0.1$ and application probability $p = 0.5$. This module has **zero learnable parameters**: it only manipulates running statistics. The key distinction from global MixStyle is per-band application, allowing different levels of style perturbation in each frequency range—directly implementing the H1 insight that high-frequency bands require more aggressive normalization.

a) Adaptive Per-Band MixStyle *(H8).:* We extend the fixed module by making α and p *learnable per band*:

$$\alpha_k = \mathrm{softplus}(\theta_k^\alpha) + \epsilon, \quad p_k = \sigma(\theta_k^p) \quad (9)$$

where $\theta_k^\alpha, \theta_k^p$ are raw parameters initialized to yield $\alpha_k = 0.1$, $p_k = 0.5$. The output is a soft blend of the original and style-mixed signals: $\hat{\mathbf{X}}_k = p_k \cdot \tilde{\mathbf{X}}_k^{\mathrm{mixed}} + (1-p_k)\cdot \mathbf{X}_k$, making the module fully differentiable via reparameterized Beta sampling. This adds only $2K = 8$ learnable parameters.

b) Spectral Content Gate (H9).: Instead of augmenting style (MixStyle), the *Spectral Content Gate* (SCG) learns to *selectively remove* it. For each sub-band k, a learnable gate $\gamma_k = \sigma(\theta_k^\gamma) \in [0, 1]$ blends between the instance-normalized (style-free) and raw (style-preserved) input:

$$\hat{\mathbf{X}}_k = \gamma_k \cdot \frac{\mathbf{X}_k - \mu_k}{\sigma_k} + (1-\gamma_k)\cdot \mathbf{X}_k \quad (10)$$

When $\gamma_k \to 1$, band k undergoes full Instance Normalization (all style removed); when $\gamma_k \to 0$, the raw signal is preserved. This adds only $K = 4$ learnable parameters. The key distinction from MixStyle is that SCG is a *deterministic, inference-time-active* module—it permanently transforms the representation rather than providing training-time augmentation.

Table II: Parameter count by model variant.

Variant	Total	Frontend	Note
A: Raw CNN (K=1)	64 586	0	Single encoder
B: Sinc FB (K=4)	249 874	8	4 encoders
C: Sinc + MixStyle	249 874	8	+0 for MixStyle
D: Sinc + MS + CS heads	172 130	8	Content/style split
E/F: UVMD (K=4)	249 910	44	Learnable decomp.
G: Sinc + Adaptive MS	249 882	16	+8 for α_k, p_k
H: UVMD + Adaptive MS	249 918	52	+8 for α_k, p_k
SCG-A: Sinc + SCG	136 974	12	+4 for γ_k
SCG-B: UVMD + SCG + Adv	172 365	48	+4 γ_k, +35K adv.

D. Stage 4: Classifier

A two-layer MLP maps $\mathbf{z} \in \mathbb{R}^{512}$ to class logits: $512 \to 256$ (ReLU, dropout $p = 0.3$) $\to 10$ (softmax).

a) Training.: The total loss is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{ov}} \cdot \mathcal{L}_{\mathrm{overlap}} \quad (11)$$

with $\lambda_{\mathrm{ov}} = 0.01$ (for UVMD only; zero for Sinc). We use Adam ($\mathrm{lr} = 10^{-3}$), ReduceLROnPlateau (factor 0.5, patience 5), and early stopping (patience 15) for up to 80 epochs.

b) Model size.: Table II summarizes parameter counts. The decomposition frontend adds negligible overhead (8–44 parameters); the dominant cost is the four per-band encoders (116 000). MixStyle adds zero parameters.

VI. Experiments and Results

A. Hypothesis Chain Overview

Table III summarizes the nine hypotheses, their status, and key results. Each hypothesis builds on the conclusions of the previous one, forming a coherent investigative chain. p-values are corrected via the Holm–Bonferroni procedure within predefined hypothesis families (see section III-D).

B. H2: Frequency Decomposition vs. Raw Signal

Table IV shows that $K = 4$ decomposition consistently outperforms raw-signal processing. Both Sinc filterbank and UVMD yield +2.1–2.3 pp F1 improvement with medium effect sizes ($d \approx 0.67$–0.68). Within Family A (decomposition, $k = 3$), both comparisons reach significance after Holm correction ($p_{\mathrm{Holm}} < 0.02$), and the controlled H6 ablation (section VI-F) independently confirms the effect ($p_{\mathrm{Holm}} < 0.003$). Sinc and UVMD are statistically indistinguishable ($\Delta = +0.24$ pp, $p = 0.50$), motivating the use of the simpler Sinc frontend in subsequent experiments.

Note on experimental series. H2–H5 were conducted with batch_size = 64; H6–H7 with batch_size = 512. This difference affects absolute F1 values (see section VII-F) and means that results from different series should not be compared directly across tables.

C. H3: Per-Band Style Normalization

Table V reveals an asymmetry between soft and hard normalization. Per-band MixStyle yields +1.67 pp with medium effect size ($d = 0.54$), but does not reach significance after Holm correction within Family B ($p_{\mathrm{Holm}} = 0.08$; $p_{\mathrm{uncorr.}} = 0.04$); we treat this as exploratory evidence. In

Table III: Hypothesis chain summary. NinaPro DB2 E1, 20 subjects, strict LOSO. p_{Holm}: Holm-corrected within hypothesis family. Status: **C** = confirmatory ($p_{\text{Holm}} < 0.05$), **E** = exploratory ($p_{\text{uncorr.}} < 0.05$ only), **R** = rejected, **D** = descriptive. Family labels: A=decomposition, B=soft style, C=hard norm, D=disentanglement, E=SCG-Net. †H2–H5: batch=64; H6–H9: batch=512 (see section VII-F).

#	Hypothesis	Key Question	Family	Status	Best ΔF1	$p_{\text{uncorr.}}$	p_{Holm}
H1	Spectral variability	Is inter-subject variability frequency-dependent?	—	**D**	—	—	—
H2†	Decomposition	Does freq. decomposition beat raw signal?	A	**C**	+2.3 pp	0.007	<0.02
H3†	Style normalization	Does per-band MIXSTYLE help?	B	**E**	+1.7 pp	0.04	0.08
		Does Instance Norm help?	C	**R (C)**	−7.0 pp	<0.001	<0.002
H4†	Disentanglement	Does explicit content/style separation help?	D	**R (C)**	−2.6 pp	<0.001	<0.004
H5†	Architecture	Do $K > 4$, SE-attention, or DRO help?	—	**R**	−0.3 pp	>0.13	—
H6	Unified ablation	Decomposition contribution (A→B)	A	**C**	+3.7 pp	<0.001	<0.003
		MIXSTYLE contribution (B→C)	B	**E**	+1.1 pp	0.07	0.07
H7	UVMD + MIXSTYLE	Is MIXSTYLE frontend-agnostic?	B	**C**	+1.8 pp	0.003	<0.01
H8	Adaptive MIXSTYLE	Does the network learn the H1 gradient?	—	**D**‡	+0.97 pp	0.064	—
H9	SCG-Net	Does per-band gate outperform uniform IN?	E	**C**	+6.86 pp	<0.001	<0.001

‡H8 is descriptive: primary evidence is the learned parameter gradient (Spearman $\rho = 0.78$, $p < 10^{-16}$), not ΔF1.

Table IV: H2: Decomposition comparison (20 subjects, LOSO).

Frontend	F1 (%)	Δ vs. raw	d
None (raw)	34.90 ± 8.16	—	—
Sinc FB (K=4)	37.00 ± 7.56	+2.10	0.67
UVMD (K=4)	37.24 ± 7.75	+2.34	0.68

Table V: H3: Style normalization comparison (20 subjects, LOSO). Baseline: Sinc FB (K=4), no normalization.

Normalization	F1 (%)	Δ	d
None (baseline)	36.28 ± 7.81	—	—
Per-band MIXSTYLE	**37.95 ± 7.22**	+1.67	0.54
Adaptive IN (γ_k)	36.90 ± 6.98	+0.62	0.24
Global IN	29.29 ± 5.30	−6.99	−1.13
Per-band IN	27.75 ± 4.29	−8.53	−1.36

contrast, hard Instance Normalization (Family C) causes large, statistically confirmed degradation: global IN loses −7.0 pp ($p_{\text{Holm}} < 0.002$, $d = -1.13$); per-band IN is worse at −8.5 pp ($p_{\text{Holm}} < 0.002$, $d = -1.36$).

The adaptive IN variant, which learns a per-band normalization strength γ_k, provides a useful diagnostic: it converges to $\gamma_1 = 0.027$ (low-frequency) versus $\gamma_4 = 0.38$ (high-frequency)—a 14:1 ratio consistent with the H1 observation that high-frequency bands carry more subject-specific variation.

D. H4: Content–Style Disentanglement

Starting from the H3 configuration with content/style projection heads (Sinc FB + MIXSTYLE + CS heads, F1 = 39.19%; note this differs from the MIXSTYLE-only result of 37.95% in table V because H4 uses a modified architecture with separate content and style encoders), we test whether explicit disentanglement losses can improve generalization (table VI). All auxiliary losses *degrade* performance:

Table VI: H4: Effect of auxiliary disentanglement losses (20 subjects). Baseline: Sinc FB + MIXSTYLE (no auxiliary losses).

Auxiliary Loss	F1 (%)	Δ	d	p_{Holm}
None (baseline)	**39.19 ± 6.34**	—	—	—
+ Contrastive	38.99 ± 5.95	−0.19	−0.14	1.00
+ Adversarial (GRL)	36.60 ± 5.99	−2.58	−2.16	<0.004
+ MI minimization	35.29 ± 7.28	−3.89	−0.97	<0.01
+ All combined	34.89 ± 7.18	−4.30	−1.04	<0.01

The adversarial loss (gradient reversal layer) worsens every subject (0/20 improved, $d = -2.16$, $p_{\text{Holm}} < 0.004$). This suggests that, in our setting, gesture content and subject style share sufficient information that forcing their separation degrades the representation. *Implicit* style robustness via MIXSTYLE appears preferable to *explicit* content–style separation, at least under this architecture and training regime.

E. H5: Architecture Optimization

We test whether increasing model capacity ($K = 6$, $K = 8$), adding Squeeze-and-Excitation attention, or GroupDRO loss can improve upon the $K = 4$ baseline. Friedman test across six variants yields $\chi^2 = 2.94$, $p = 0.71$—no significant differences. The best variant ($K = 4$, no additions, F1 = 39.75%) remains the simplest. Within the range of architectures tested, additional complexity does not improve performance; however, this does not rule out gains from fundamentally different architectures or training strategies.

F. H6: Unified Ablation (Component Attribution)

To isolate individual component contributions, we fix a single backbone architecture and vary one component per step (table VII). Note that H6–H7 use batch_size = 512 (vs. 64 in H2–H5), which affects absolute F1 values and limits direct cross-table comparison (see section VII-F).

Key findings:

Table VII: H6: Unified ablation (20 subjects, batch=512, LOSO). Each step adds exactly one component. Statistical tests: Wilcoxon with Holm correction within hypothesis families (A, B).

	Components	F1 (%)	IQR	Δ_{step}	d	p_{Holm}	↑ / ↓
A	Raw CNN	32.40 ± 5.91	8.9	—	—	—	—
B	+ Sinc FB	36.08 ± 7.29	10.4	+3.68	1.25	<0.003	18/2
C	+ MixStyle	**37.19** ± **6.21**	8.5	+1.11	0.42	0.07*	18/2
D	+ CS heads	36.37 ± 6.07	8.8	−0.82	−0.62	0.34	5/15

*Exploratory: does not reach significance within Family B.

Table VIII: H7: UVMD + MixStyle (20 subjects, batch=512, LOSO; Variant F replicated at batch=64 with $p = 0.67$). E→F: $p_{\text{Holm}} < 0.01$ (Family B, confirmatory).

	Frontend + Style	F1 (%)	Δ	d	$p_{\text{uncorr.}}$
E	UVMD	35.79 ± 6.67	—	—	—
F	UVMD + MixStyle	**37.58** ± **6.39**	+1.79	0.84	0.003
	Cross-frontend comparison:				
B	Sinc FB	36.08 ± 7.29	—	−0.16	0.78
E	UVMD	35.79 ± 6.67			
C	Sinc + MS	37.19 ± 6.21	+0.39	0.25	0.43
F	UVMD + MS	37.58 ± 6.39			

- **Decomposition** (A→B): +3.68 pp, $d = 1.25$ (large), $p_{\text{Holm}} < 0.003$ (Family A). Confirmatory.
- **MixStyle** (B→C): +1.11 pp, $d = 0.42$ (small–medium), $p_{\text{uncorr.}} = 0.07$. Consistent direction (18/20 subjects improve) but does not reach significance. Exploratory.
- **Content/Style heads** (C→D): −0.82 pp, harmful (15/20 worsen).

The best-performing variant within this ablation is **Variant C** (Sinc filterbank + per-band MixStyle), though the marginal contribution of MixStyle over decomposition alone is not individually confirmed after correction.

G. H7: Per-Band MixStyle on Learnable Decomposition

Does MixStyle help only with fixed decomposition, or does it generalize to learnable frontends? We test on UVMD (table VIII).

MixStyle adds +1.79 pp on UVMD ($d = 0.84$, large effect size). Within Family B ($k = 3$), H7 is the smallest p-value (0.003) and passes the Holm step ($p_{\text{Holm}} < 0.01$), making this a **confirmatory** result: per-band MixStyle significantly improves the learnable frontend. However, the sequential Holm procedure stops at the next comparison (H3, $p_{\text{uncorr.}} = 0.04 > \alpha/2 = 0.025$), so the effect on the fixed frontend remains exploratory. The effect is numerically larger on learnable decomposition (+1.79 pp) than on fixed (+1.11 pp), possibly because UVMD's adaptive frequency placement creates sub-bands where style statistics are more informative, though this difference is not tested.

Sinc and UVMD frontends remain statistically indistinguishable ($p = 0.78$ without MixStyle, $p = 0.43$ with), with UVMD better for 10/20 subjects and Sinc for 10/20. As a batch-size robustness check, Variant F was replicated at batch_size = 64: 37.69% ± 6.85 (paired Wilcoxon $p = 0.67$ vs. the bs=512 result), confirming that the finding is not an artifact of batch size (see section VII-F).

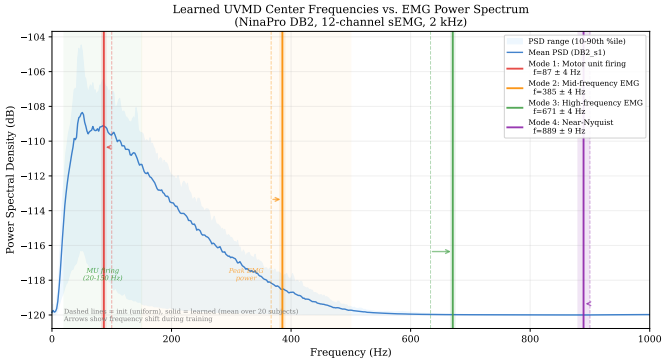

Figure 4: Learned UVMD center frequencies (solid lines, mean over 20 subjects) overlaid on EMG PSD (representative subject). Dashed lines: initial (uniform) frequencies. Arrows show adaptation direction.

Table IX: H8: Adaptive Per-Band MixStyle (20 subjects, batch=512, LOSO). G/H: learnable α_k, p_k; I/J: reversed-gradient ablation (learned values in wrong band order).

	Frontend + Style	F1 (%)	Acc (%)	ΔF1 vs. rev.
G	Sinc + **Adaptive MS**	**35.56** ± **7.05**	**37.23** ± **6.59**	+0.24
I	Sinc + Reversed MS	35.32 ± 7.32	36.97 ± 6.62	—
H	UVMD + **Adaptive MS**	**34.42** ± **6.91**	**36.79** ± **6.29**	+0.97
J	UVMD + Reversed MS	33.46 ± 6.40	35.53 ± 6.02	—

a) Learned UVMD *frequencies.:* Figure 4 shows the learned center frequencies overlaid on the EMG power spectrum. Starting from uniform initialization at [100, 367, 633, 900] Hz, UVMD converges to [87±4, 385±4, 671±4, 889±9] Hz across all 20 subjects (std < 10 Hz), indicating a *universal*, subject-independent frequency decomposition. Mode 1 shifts toward the motor-unit firing band; Mode 2 centers on peak EMG power.

H. H8: Adaptive Per-Band MixStyle

Does the network *learn* the frequency-dependent style augmentation implied by H1, or is the fixed $\alpha = 0.1$, $p = 0.5$ already optimal? We replace the fixed per-band MixStyle with the adaptive variant (eq. (9)) on both Sinc (variant G) and UVMD (variant H) frontends. The MixStyle parameters receive $10\times$ the base learning rate to ensure convergence within 80 epochs. As an ablation, we also test *reversed-gradient* variants (I, J) that use the learned α_k, p_k values in *reversed band order* (band 4's values assigned to band 1 and vice versa), to verify that the specific band assignment is causally important.

Table IX shows that the correct band ordering outperforms the reversed ordering on both frontends. The effect is stronger on UVMD (+0.97 pp, Wilcoxon $p = 0.064$, 15/20 subject wins) than on Sinc (+0.24 pp, $p = 0.67$). The primary finding, however, is not the F1 difference but the *learned parameter gradient* (table X).

Table X: Learned Adaptive MIXSTYLE parameters (mean ± std across 20 LOSO folds). Init: $\alpha_k = 0.1$, $p_k = 0.5$ for all bands.

Band	G (Sinc)		H (UVMD)	
k	α_k	p_k	α_k	p_k
1 (low)	1.62 ± 0.10	<0.001	1.30 ± 0.30	0.018 ± 0.027
2	0.89 ± 0.11	<0.001	1.08 ± 0.24	0.014 ± 0.022
3	**7.80 ± 1.24**	0.010 ± 0.002	**3.04 ± 1.45**	0.036 ± 0.029
4 (high)	**16.28 ± 2.43**	**0.040 ± 0.006**	**2.62 ± 1.36**	**0.046 ± 0.051**
Ratio $k=4/k=1$	**10.0**×	>100×	2.0×	2.5×
Spearman ρ	0.775 ($p < 10^{-16}$)		0.347 ($p = 0.002$)	

Both frontends exhibit a *monotonic gradient* in the learned parameters:

- α_k **increases with frequency**: from 1.62 (band 1) to 16.28 (band 4) on Sinc (**10**×), and from 1.30 to 2.62 on UVMD (2×). Higher α produces a more concentrated Beta distribution, meaning style mixing becomes less random in high-frequency bands.
- p_k **increases with frequency**: from <0.001 to 0.040 on Sinc, from 0.018 to 0.046 on UVMD. The network effectively disables style mixing in low-frequency bands and enables it selectively in high-frequency bands.

Spearman rank correlation between band index and α_k is $\rho = 0.775$ ($p < 10^{-16}$) on Sinc and $\rho = 0.347$ ($p = 0.002$) on UVMD. This pattern is consistent across all 20 LOSO folds and both frontends.

a) Interpretive caveats.: The H8 evidence is *descriptive*, not confirmatory: the Spearman correlation quantifies a pattern in learned parameters, not a controlled manipulation. The parameters are optimized jointly with all other weights; we cannot rule out that the monotonic gradient is an indirect consequence of other architectural or data properties rather than a direct encoding of the H1 variability structure. The reversed-gradient ablation provides a partial causal control (correct ordering outperforms reversed on UVMD, $p_{\text{uncorr.}} = 0.064$, 15/20 wins), but the effect is small (+0.97 pp) and does not reach significance. We interpret H8 as *suggestive* evidence that complements, but does not independently confirm, the H1 observation.

I. H9: Spectral Content Gate Network (SCG-Net)

Can the H1 insight be encoded directly as an *architectural prior* rather than a training-time augmentation? SCG-Net replaces MixStyle with the Spectral Content Gate (eq. (10)): a per-band learnable gate γ_k that blends between instance-normalized (style-free) and raw input, adding only $K = 4$ parameters.

Table XI shows that SCG-Net (variant C) achieves 35.65% F1, significantly outperforming uniform Instance Normalization (variant D, 28.79%) by +6.86 pp ($p_{\text{Holm}} < 0.001$, Family E, 18/20 subject wins). The learned gate values (table XII) reveal a near-binary partition: $\gamma_{1\text{–}3} < 0.001$ (preserve all style in low-frequency bands) and $\gamma_4 = 0.998$ (remove style in the highest band). This pattern is consistent with the H1 observation, though it is learned within the same architectural framework rather than constituting a fully independent validation.

Table XI: H9: SCG-Net results (20 subjects, batch=512, LOSO). Variant C is the main architecture; A adds a subject adversary; D uses fixed $\gamma_k = 1$ (uniform Instance Norm, ablation).

	Configuration	F1 (%)	Acc (%)	ΔF1 vs. D
SCG-C	Sinc + **SCG** (learned γ_k)	**35.65 ± 6.60**	**37.39 ± 6.07**	+6.86
SCG-A	Sinc + SCG + Adversary	34.75 ± 5.92	36.17 ± 5.82	+5.97
SCG-B	UVMD + SCG + Adversary	34.56 ± 5.97	36.01 ± 5.65	+5.77
SCG-D	Sinc + full IN ($\gamma_k = 1$ fixed)	28.79 ± 4.09	29.32 ± 3.94	—

Table XII: Learned SCG gate values γ_k (mean ± std across 20 LOSO folds). Init: $\gamma_k = 0.5$ for all bands.

Band k	SCG-C (Sinc)	SCG-A (Sinc + Adv)
1 (low)	<0.001	<0.001
2	<0.001	<0.001
3	<0.001	0.001 ± 0.001
4 (high)	**0.998 ± 0.002**	**0.910 ± 0.184**
Spearman ρ	0.433 ($p < 10^{-4}$)	0.774 ($p < 10^{-16}$)

The subject adversary (variant A) slightly reduces performance compared to SCG without adversary (C), consistent with the H4 finding that adversarial disentanglement can harm sEMG classification. Wilcoxon test confirms this difference is significant ($p = 0.048$, 12/20 wins for C over A).

J. Comparison with Published Baselines

Table XIII compares our results with the published LOSO baselines from Golovan and Zvereva [9]. Our best variant (UVMD+MIXSTYLE) yields 37.7% F1 (37.69% at bs=64, 37.58% at bs=512), compared with the previous best (CNN/LSTM BiGRU, 33.5%). Note that the comparison is between different architectures and training regimes; no paired statistical test is possible since the baselines were trained separately.

K. Who Benefits from MIXSTYLE?

Figure 5 shows that MIXSTYLE's benefit correlates negatively with baseline F1 for the Sinc frontend ($r = -0.56$, $p = 0.01$): weaker subjects gain more, while strong subjects see diminishing returns. This suggests that MIXSTYLE acts as a regularizer, primarily helping subjects whose signals deviate most from the training distribution.

VII. ANALYSIS AND DISCUSSION

A. Decomposition as the Primary Confirmed Effect

The H6 ablation shows decomposition (+3.68 pp, $d = 1.25$, $p_{\text{Holm}} < 0.003$) contributes roughly three times more than MIXSTYLE (+1.11 pp, $d = 0.42$, $p_{\text{uncorr.}} = 0.07$). This ratio is consistent with the H1 observation that frequency structure is a major axis of subject-specific variation. By isolating sub-bands, the per-band encoders can specialize: low-frequency encoders focus on motor-unit patterns (high Fisher ratio), while high-frequency encoders can learn to suppress noise (high normalized CV). The MIXSTYLE effect on fixed

Table XIII: Comparison with published baselines (LOSO, 20 subjects, NinaPro DB2 E1). Published results from [9]. †Replication run at batch=64.

Method	Features	F1 (%)
CNN/LSTM BiGRU [9]	Raw EMG	33.5 ± 8.0
Simple CNN [9]	Raw EMG	33.3 ± 9.1
SVM-RBF [9]	TD Feat.	32.6 ± 7.7
SVM-Linear [9]	TD Feat.	32.5 ± 9.4
Ours: Sinc + MIXSTYLE (C)	Raw EMG	37.19 ± 6.21
Ours: UVMD + MIXSTYLE (F)	**Raw EMG**	**37.69 ± 6.85**†

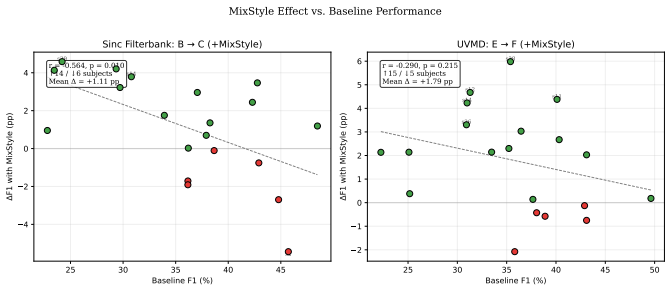

Figure 5: Per-subject MIXSTYLE F1 gain vs. baseline F1. Left: Sinc FB (B→C), significant negative correlation. Right: UVMD (E→F), similar trend.

Sinc decomposition (H3, H6 B→C) remains exploratory: it does not pass Holm correction within Family B. On the learnable UVMD frontend (H7), the effect is confirmatory ($p_{\text{Holm}} < 0.01$, $d = 0.84$), suggesting that the benefit may depend on the frontend type or on the larger effect size achievable with adaptive decomposition. We caution against interpreting MIXSTYLE as universally beneficial for sEMG; the evidence is mixed and frontend-dependent.

B. H8: Learned Parameter Gradient as Descriptive Evidence

The H8 experiment shows that a network optimized purely for classification loss learns α_k and p_k that increase monotonically from low to high frequency bands, mirroring the H1 observation. The Sinc frontend exhibits a 10× ratio ($\alpha_4/\alpha_1 = 16.3/1.6$), with Spearman $\rho = 0.775$ ($p < 10^{-16}$).

However, this evidence is *descriptive*: the correlation between learned parameters and band index is an observation about the optimization outcome, not a controlled experiment. The reversed-gradient ablation (UVMD: -0.97 pp, $p_{\text{uncorr.}} = 0.064$, 15/20 wins) provides a partial causal control, but the effect is small and non-significant. We interpret the parameter gradient as *suggestive* of a frequency-dependent structure, consistent with H1 but not independently confirming it through controlled manipulation.

The overall reduction in p_k from initialization (0.5 to <0.001–0.04) is itself noteworthy: the network determines that the commonly used $p = 0.5$ over-regularizes, and that low-frequency bands should receive essentially no style mixing. If replicated on additional datasets, this finding could inform practical design choices for per-band augmentation.

C. SCG-Net: Per-Band Style Removal

The progression H1→H8→H9 traces a methodological arc: *observe* the frequency-dependent variability (H1), examine whether learned augmentation parameters reflect it (H8, descriptive), then test whether encoding it as an architectural prior helps (H9). The SCG gate differs from MixStyle: instead of training-time augmentation, it *permanently removes* subject-specific variation from selected bands at both training and inference time. The near-binary gate values ($\gamma_4 = 0.998$, $\gamma_{1\text{–}3} < 0.001$) suggest that the optimal strategy is a hard frequency-dependent partition, though this conclusion is conditioned on our specific architecture and dataset.

The +6.86 pp advantage of SCG over uniform IN (table XI) quantifies the cost of frequency-agnostic normalization. However, SCG-Net (35.65% F1) does not exceed the MixStyle-based pipeline (37.19–37.69% F1), indicating that selective style *removal* is less effective than style *augmentation* for overall generalization. SCG-Net's primary value in this study is as a diagnostic tool that confirms the binary content/style partition across frequency bands, rather than as a state-of-the-art architecture.

D. Why Hard Normalization Fails

Instance Normalization removes *all* first- and second-order statistics, including channel-wise amplitude patterns that carry gesture information (which channels activate, and how strongly). In contrast, MIXSTYLE only *interpolates* between subjects' statistics, preserving the overall distribution structure while increasing training-time diversity. The H8 experiment is consistent with this interpretation: learned parameters indicate minimal style mixing in low-frequency bands and selective mixing in high-frequency bands, though this is descriptive evidence (see section VI-H). SCG-Net (H9) provides complementary evidence: when given a per-band gate ($\gamma_k \in [0, 1]$), the network removes style only from the highest band ($\gamma_4 \approx 1$) and preserves it in all others.

E. Why Adversarial Disentanglement Harms Performance

Adversarial disentanglement (GRL) assumes that a representation can be split into subject-invariant "content" and subject-specific "style" components. In our experiments, this assumption appears to fail: the amplitude pattern across 12 electrodes likely encodes both the gesture (which muscles activate) *and* the subject's physiology (electrode placement, muscle mass, subcutaneous fat). Forcing orthogonality between content and style projections may destroy gesture-relevant information. The MI-minimization variant shows similar degradation.

We note that this conclusion is specific to the adversarial/MI-based disentanglement mechanisms tested in H4. Separate exploratory experiments on a reduced 5-subject subset suggest that other disentanglement approaches (e.g., channel-contrastive with Group DRO) may avoid this failure mode, but these results cannot be compared directly with the 20-subject findings.

F. Batch-Size Confound

A limitation of this study is the change in batch size between experimental series: H2–H5 used batch_size = 64, while H6 used batch_size = 512. This difference was introduced for computational efficiency in the unified ablation but has several implications:

- **Absolute F1 values differ across series.** For example, the raw CNN baseline is 34.90% in H2 (bs=64) vs. 32.40% in H6 (bs=512). The Sinc+MIXSTYLE configuration is 37.95% in H3 vs. 37.19% in H6. These are *not* directly comparable.
- **Batch size affects BatchNorm and MIXSTYLE.** Larger batches provide better BatchNorm statistics but reduce the diversity of MIXSTYLE pairings within a batch, potentially weakening the augmentation effect.
- **Component attributions from H6 are internally consistent** (same batch size throughout), but the absolute magnitudes may differ from what would be observed at batch_size = 64.

To verify that the H7 conclusion is robust to this confound, we replicated H7 Variant F (UVMD+MIXSTYLE) at batch_size = 64 (matching H2–H5). The result—37.69% ± 6.85—is not significantly different from the bs=512 run (37.58% ± 6.39; paired Wilcoxon $p = 0.67$), confirming that the UVMD+MIXSTYLE effect is batch-size–invariant. The bs=512 E→F comparison remains internally consistent and is reported in table VIII.

G. Limitations

1) **Single dataset and exercise.** All results are on NinaPro DB2 E1 (10 gestures, 12 channels, 2000 Hz). Generalization to other NinaPro databases, exercises, electrode configurations, or recording hardware is untested.
2) **Subset of subjects.** We use 20 of 40 available subjects. While chosen to span the full database range, this limits external validity.
3) **Batch-size confound.** As discussed in section VII-F, the change in batch size between H2–H5 and H6 complicates cross-table comparisons. The H7 main result has been replicated at batch_size = 64 ($p = 0.67$ vs. original), but the H6 ablation was not re-run.
4) **Mixed evidence for MIXSTYLE.** Per-band MIXSTYLE reaches confirmatory significance only on the learnable UVMD frontend (H7). On fixed Sinc, the effect is consistent but exploratory. The benefit should not be treated as established until replicated.
5) **H8 is descriptive.** The learned parameter gradient is an observational finding from parameter inspection, not a controlled comparison. The reversed-gradient ablation is small and non-significant. Conclusions about "the network learning the H1 gradient" should be interpreted cautiously.
6) **SCG-Net comparison is narrow.** SCG-Net is compared against uniform IN (an intentionally weak baseline) and MixStyle variants, but not against other domain generalization architectures. Its value in this study is primarily diagnostic.
7) **No real-time evaluation.** All experiments are offline; latency and computational constraints of embedded deployment are not addressed.
8) **Novelty.** Frequency decomposition and MIXSTYLE are established methods; our contribution is their systematic per-band evaluation for cross-subject sEMG. The Spectral Content Gate is a novel but simple architectural element whose broader applicability remains untested.

VIII. CONCLUSION

We presented a hypothesis-driven investigation (H1–H9) of frequency-aware processing for cross-subject sEMG gesture recognition on NinaPro DB2 (20 subjects, 10 gestures, LOSO).

a) Confirmatory findings ($p_{\mathrm{Holm}} < 0.05$).:

- $K = 4$ frequency decomposition adds +3.7 pp F1 ($d = 1.25$) over a raw-signal baseline (H6; also confirmed in H2).
- Per-band MIXSTYLE on learnable UVMD decomposition adds +1.8 pp ($d = 0.84$, H7). On fixed Sinc decomposition, the effect is consistent in direction but remains exploratory.
- Hard Instance Normalization is strongly harmful (-7 to -8.5 pp).
- Adversarial and MI-based disentanglement degrade performance (-2.6 to -4.3 pp) under the architectures tested.
- SCG-Net's per-band gate outperforms uniform Instance Normalization by +6.86 pp ($p_{\mathrm{Holm}} < 0.001$).

b) Descriptive and exploratory findings.:

- Per-band MIXSTYLE on fixed Sinc: +1.1–1.7 pp, consistent direction but not confirmed after correction.
- Adaptive Per-Band MIXSTYLE (H8) learns parameters correlated with the H1 gradient ($\alpha_4/\alpha_1 = 10\times$, $\rho = 0.78$), but this is descriptive evidence from parameter inspection, not a controlled comparison.
- SCG-Net learns a near-binary gate partition ($\gamma_4 = 0.998$, $\gamma_{1\text{–}3} < 0.001$), consistent with H1 but based on the same architectural framework.

The combined pipeline achieves 37.7% macro-F1 (UVMD+MIXSTYLE, replicated at two batch sizes), compared with the published baseline of 33.5%. These results are obtained on a single dataset with a subset of subjects and should be validated on other sEMG benchmarks and electrode configurations before broader conclusions are drawn. Within these limitations, our findings suggest that frequency-dependent inter-subject variability is a significant structural property of sEMG that can be addressed through per-band processing, with decomposition providing the largest confirmed benefit.

REFERENCES

[1] M. Atzori *et al.*, "Deep learning with convolutional neural networks applied to electromyography data: A resource for the classification of movements for prosthetic hands," *Front. Neurorobot.*, vol. 10, 2016.

[2] M. Atzori *et al.*, "Electromyography data for non-invasive naturally-controlled robotic hand prostheses," *Sci. Data*, vol. 1, p. 140053, 2014.

[3] D. Farina *et al.*, "The extraction of neural information from the surface EMG for the control of upper-limb prostheses: emerging avenues and challenges," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 4, pp. 797–809, 2014.

[4] T. S. Saponas *et al.*, "Enabling always-available input with muscle-computer interfaces," in *Proc. UIST*, 2009.

[5] E. Campbell *et al.*, "Current trends and confounding factors in myoelectric control: Limb position and contraction intensity," *Sensors*, vol. 20, no. 6, 2020.

[6] U. Côté-Allard *et al.*, "Deep learning for electromyographic hand gesture signal classification using transfer learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 4, pp. 760–771, 2019.

[7] Y. Du *et al.*, "Surface EMG-based inter-session gesture recognition enhanced by deep domain adaptation," *Sensors*, vol. 17, no. 3, 2017.

[8] A. Phinyomark *et al.*, "Feature reduction and selection for EMG signal classification," *Expert Syst. Appl.*, vol. 39, no. 8, pp. 7420–7431, 2012.

[9] K. Golovan and I. Zvereva, "A reproducible benchmark for zero-shot cross-subject hand gesture recognition using sEMG signals," 2025.

[10] K. Dragomiretskiy and D. Zosso, "Variational mode decomposition," *IEEE Trans. Signal Process.*, vol. 62, no. 3, pp. 531–544, 2014.

[11] M. Ravanelli and Y. Bengio, "Speaker recognition from raw waveform with SincNet," in *Proc. SLT*, 2018.

[12] K. Zhou *et al.*, "Domain generalization with MixStyle," in *Proc. ICLR*, 2021.

[13] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, 1993.

[14] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, 2003.

[15] E. Scheme and K. Englehart, "Electromyogram pattern recognition for control of powered upper-limb prostheses: State of the art and challenges for clinical use," *J. Rehabil. Res. Dev.*, vol. 48, no. 6, pp. 643–659, 2011.

[16] S. Pizzolato *et al.*, "Comparison of six electromyography acquisition setups on hand movement classification tasks," *PLoS ONE*, vol. 12, no. 10, e0186132, 2017.

[17] X. Zhai *et al.*, "Self-recalibrating surface EMG pattern recognition for neuroprosthesis control based on convolutional neural network," *Front. Neurosci.*, vol. 11, 379, 2017.

[18] W. Wei *et al.*, "A multi-stream convolutional neural network for sEMG-based gesture recognition in muscle-computer interface," *Pattern Recognit. Lett.*, vol. 119, pp. 131–138, 2019.

[19] I. Ketykó, F. Kovács, and K. Z. Varga, "Domain adaptation for sEMG-based gesture recognition with recurrent neural networks," in *Proc. IJCNN*, 2019.

[20] N. E. Huang *et al.*, "The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis," *Proc. R. Soc. Lond. A*, vol. 454, pp. 903–995, 1998.

[21] N. Rehman and D. P. Mandic, "Multivariate empirical mode decomposition," *Proc. R. Soc. A*, vol. 466, pp. 1291–1302, 2010.

[22] K. Englehart *et al.*, "Classification of the myoelectric signal using time-frequency based representations," *Med. Biol. Eng. Comput.*, vol. 37, no. 4, pp. 526–532, 1999.

[23] A. Phinyomark and E. Scheme, "EMG pattern recognition in the era of big data and deep learning," *Big Data Cogn. Comput.*, vol. 2, no. 3, 21, 2018.

[24] D. Borra *et al.*, "Interpretable and lightweight convolutional neural network for EEG decoding: Application to movement execution and imagination," *Neural Netw.*, vol. 129, pp. 55–74, 2020.

[25] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015.

[26] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: The missing ingredient for fast stylization," *arXiv:1607.08022*, 2016.

[27] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proc. ICCV*, 2017.

[28] S. Sagawa *et al.*, "Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization," in *Proc. ICLR*, 2020.

[29] M. Arjovsky *et al.*, "Invariant risk minimization," *arXiv:1907.02893*, 2019.

[30] Y. Ganin *et al.*, "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.

[31] J. Wang *et al.*, "Generalizing to unseen domains: A survey on domain generalization," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8052–8072, 2023.

[32] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian J. Statistics*, vol. 6, no. 2, pp. 65–70, 1979.